TEAM 07 · INTERSYSTEMS READY 2026

SignBridge

Pre-Visit Triage for Deaf Patients

Agent Flow & Architecture

What is SignBridge?

SignBridge prepares a **pre-visit clinical briefing** before a deaf patient connects to a doctor through a sign-language interpreter. When the appointment is booked, one MCP call triggers a multi-agent pipeline inside IRIS that produces a structured brief covering symptoms, patient history, triage urgency, and a medical glossary tailored to the patient's sign language.

The problem solved: interpreters arrive cold, doctors have no symptom context, and triage urgency is invisible until the call is already underway. SignBridge fixes all three — before the call starts.

The Six Agents

Agent	Role	Tools	Output (JSON)
Orchestrator parent agent	Plans the 9-step call sequence. Delegates to four specialists, reviews results, triggers a re-query loop when TriageScorer flags ESI ≤ 2, then calls ReportWriterAgent.	KbSearch · SqlListTables · SqlDescribe · SqlSelect · AnalyzeSymptoms · SummarizeHistory · ScoreTriage · BuildGlossary · AssembleReport	Delegates — final output is the Markdown brief
SymptomAnalyzer sub-agent	Analyses the chief complaint against the medical knowledge base. Returns likely conditions, differential diagnoses, red flags, and topic keywords for downstream agents.	KbSearch (RAG over local medical docs, 384-dim FastEmbed)	`likely_conditions · differential · red_flags · topic_keywords · citations`
HistoryAgent sub-agent	Pulls the patient's clinical history from the FHIR-style dataset-health tables: prior visits, active problems, current medications, allergies, and relevant labs.	SqlListTables · SqlDescribe · SqlSelect	`prior_visits · active_problem · current_meds · relevant_labs · context_for_doc`
TriageScorer sub-agent	Scores urgency using the Emergency Severity Index (ESI 1–5). No retrieval tools — pure clinical reasoning over the symptom and history JSON. Sets `must_re_query_kb_for_red_flags` when ESI ≤ 2 to trigger the agentic loop.	None (pure reasoning)	`esi_level · label · justification · red_flags · escalations_applied · must_re_query_kb_for_red_flags`
InterpreterBriefer sub-agent	Builds a sign-language glossary for the interpreter. Matches medical terms to ASL/BSL/PSL sign cues, sets the communication tone (calm vs urgent), and produces tips for the specific sign style requested.	KbSearch (glossary & communication docs)	`glossary[{term, plain_english sign_cue}] · topic_summary · communication_tone · communication_tips`
ReportWriterAgent sub-agent	Assembles all four JSON blocks into a single structured Markdown brief — one section for the doctor, one for the interpreter, plus sourced citations from the knowledge base.	None (composition only)	Markdown brief (the final deliverable)

Step-by-Step Flow

1. **Patient intake received**

   Streamlit sends a JSON intake object — `chief_complaint`, `duration`, `age`, `sex`, `sign_style`, `known_conditions`, `current_meds`, optional `patient_id` — to the IRIS MCP endpoint `/mcp/signbridge` via the `ProcessIntake` tool.

2. **Knowledge base search (initial)**

   The Orchestrator calls **KbSearch** with the chief complaint (topK=4). FastEmbed converts the query to a 384-dim vector and retrieves the most relevant chunks from the local SignBridge.KBVectors table. Result: evidence_v1.

3. **SymptomAnalyzer (first pass)**

   Orchestrator calls the **AnalyzeSymptoms** delegate tool with the intake and evidence_v1. A fresh SymptomAnalyzer sub-agent is spawned, runs, and returns symptoms_v1 — a structured JSON with conditions, differentials, and red flags.

4. **History lookup (if patient_id present)**

   Orchestrator calls **SqlListTables** and then 1–3 **SqlSelect** queries to retrieve visit history, medications, and labs from the dataset-health FHIR tables. Results passed to the HistoryAgent delegate.

5. **HistoryAgent**

   Spawned with the intake and SQL results. Returns history JSON summarising prior visits, active problems, current meds, allergies, and a short context note for the doctor.

6. **TriageScorer**

   Receives intake + symptoms_v1 + history. Applies the ESI 1–5 rubric purely from reasoning (no tools). Returns triage JSON including ESI level, justification, red flags list, and the `must_re_query_kb_for_red_flags` flag.

7. **⚡ Agentic re-query loop (ESI 1–2 only)**

   If ESI ≤ 2 and the flag is set, the Orchestrator calls **KbSearch** again with `redFlagOnly=true` to surface high-urgency evidence, then re-runs **AnalyzeSymptoms** with the combined evidence set. The result (symptoms_final) replaces symptoms_v1 for the brief. For routine cases (ESI 3–5) this loop is skipped entirely.

8. **InterpreterBriefer**

   KbSearch for glossary + communication docs → BuildGlossary delegate. Returns a per-term sign-cue glossary, a communication tone rating, and interpreter tips tuned to the patient's sign style (ASL / BSL / PSL / …).

9. **ReportWriterAgent**

   AssembleReport tool packages intake + symptoms_final + history + triage + glossary into a Markdown brief. ReportWriterAgent composes two sections: one for the **doctor** (differential, urgency, history context) and one for the **interpreter** (glossary, communication tips, sign cues). The Orchestrator returns the brief to ProcessIntake, which returns it to Streamlit.

The Agentic Loop — Why This Is Not a Pipeline

> A chained pipeline always executes the same steps in the same order. SignBridge's Orchestrator **reviews** the TriageScorer output and makes a conditional decision: if the patient is flagged ESI 1 or 2 (immediate / emergent), it goes back to the knowledge base with a red-flag filter and re-runs the symptom analysis before composing the brief. This loop is driven by the Orchestrator's INSTRUCTIONS XData — not hard-coded in ObjectScript — so it is genuine LLM-mediated planning, not a fixed branch.
>
> Demo prompt that triggers it: "BSL speaker, age 58, diabetic on metformin, chest pain radiating to left arm for 2 hours." TriageScorer returns ESI 1 → loop fires → cardiac red-flag evidence retrieved → SymptomAnalyzer adds aortic dissection and STEMI to the differential → brief surfaces emergency interpreter preparation notes.

Technology Stack

Layer	Technology	Detail
Agent framework	IRIS AI Hub 2026.2	`%AI.Agent`, `%AI.Tool`, `%AI.ToolSet`
LLM	OpenAI gpt-5-nano	One provider key; all 6 agents share the same model
RAG / embeddings	FastEmbed (local)	384-dim, no API key, `SignBridge.KBVectors` table
Patient data	dataset-health (FHIR)	IRIS SQL, `Health.*` tables, read-only guard
MCP transport	iris-mcp-server (Rust)	HTTP+SSE on :8080, WgProto to IRIS :1972
UI	Streamlit	Two-pane: intake form + live brief, :8501
Container	IRIS-for-Health 2026.2.0AI	Docker, IRISAPP namespace

Knowledge Base Corpus (8 documents)

- **chest_pain_triage.md** — Red flags, differential, cardiac vs non-cardiac signs
- **diabetes_complications.md** — DKA, hypoglycaemia, neuropathy, renal involvement
- **hypertension.md** — Hypertensive urgency vs emergency, BP thresholds
- **asthma_exacerbation.md** — Severity grading, peak flow, albuterol dosing
- **common_medications.md** — 20+ drugs: class, indication, common side-effects
- **esi_reference.md** — ESI 1–5 rubric with clinical examples
- **deaf_communication_tips.md** — Doctor/interpreter communication guidance

- **sign_language_medical_glossary.md** — 30 terms in ASL / BSL / PSL with sign cues